

**DUW10042**

# **OCCUPATIONAL SAFETY & HEALTH FOR ENGINEERING**

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## **FULL LECTURE NOTES**

Polytechnic-aligned | Malaysian OSH legislation updated to August 2026

### **IMPORTANT**

These are independently prepared study notes aligned to the publicly available DUW10042 course description and current DOSH legislation. They are not an official lecturer handout or a reproduction of restricted POLYCC notes.

*Designed for lecture revision, tutorial preparation and examinations*

# How to Use These Notes

The course is built around the idea that safety is an engineering and management responsibility, not simply the use of personal protective equipment (PPE). Read each chapter in sequence, then use the exam-focus boxes and practice questions for revision.

| Chapter | Main area                          | What you should be able to do                                                                                   |
|---------|------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1       | Introduction & OSH Legislation     | Explain OSH concepts, legal duties, employee rights and the current Malaysian legal framework.                  |
| 2       | OSH Management                     | Explain policy, planning, responsibilities, committee/coordinator functions, training and emergency management. |
| 3       | Incident Prevention                | Differentiate accident/incident/near miss, explain causation, investigation, reporting and corrective action.   |
| 4       | Chemical Safety & HIRARC           | Identify hazards, interpret SDS/labels, assess risk, complete HIRARC and choose controls.                       |
| 5       | Fire Safety & First Aid            | Explain fire science, fire classes, extinguishers, evacuation and basic first-response priorities.              |
| 6       | Workplace Environment & Ergonomics | Recognise physical, environmental, ergonomic and psychosocial risks and propose controls.                       |

## 2026 LEGAL UPDATE

The core law is the Occupational Safety and Health Act 1994 (Act 514), incorporating the Occupational Safety and Health (Amendment) Act 2022, which came into operation on 1 June 2024. Major current features include a legal duty to conduct and implement risk assessment (s.18B), OSH Coordinator requirements for many workplaces with 5 or more employees (s.29A), employee right to remove themselves from imminent danger in defined circumstances (s.26A), wider application of the Act, and substantially higher penalties. DOSH also lists new subsidiary legislation including the 2024 Plant Requiring Certificate of Fitness Regulations, 2024 Construction Work (Design and Management) Regulations, and 2025 Special Scheme of Inspection Regulations.

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- Chapter 1 - Introduction to Occupational Safety and Health & Legislation
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## CHAPTER 1

# INTRODUCTION TO OSH & LEGISLATION

From safety concepts to the current Malaysian legal framework

## 1.1 What is Occupational Safety and Health?

Occupational Safety and Health (OSH) is the systematic protection of people from work-related injury, ill health, disease and unsafe conditions. It combines engineering controls, management systems, human factors, occupational hygiene, ergonomics, emergency management and legal compliance.

| Term                 | Meaning                                                                     | Example                                                              |
|----------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------|
| Safety               | Control of conditions that can cause injury, damage or loss.                | Machine guarding prevents contact with rotating parts.               |
| Occupational health  | Protection and promotion of physical, mental and social well-being at work. | Noise control prevents occupational hearing loss.                    |
| Hazard               | Source, situation or act with potential to cause harm.                      | Exposed electrical conductor.                                        |
| Risk                 | Combination of likelihood and severity of harm.                             | Chance and consequence of electric shock from the exposed conductor. |
| Incident             | Work-related event in which injury/ill health could or does occur.          | Forklift almost strikes a pedestrian.                                |
| Near miss            | Incident with no injury/damage but clear potential for harm.                | Tool falls from height but misses workers.                           |
| Occupational disease | Disease caused or aggravated by workplace exposure.                         | Noise-induced hearing loss, occupational dermatitis.                 |

## 1.2 Why OSH Matters in Engineering

- **Human reason:** Every person should return home without injury or work-related illness.
- **Legal reason:** Employers, principals, designers, manufacturers, suppliers and employees have statutory duties.
- **Economic reason:** Incidents create direct costs (treatment, repairs, compensation) and indirect costs (downtime, investigation, lost productivity, retraining, reputation).
- **Engineering reason:** Safe systems improve reliability, maintainability and process control.
- **Professional reason:** Engineers influence design decisions that can eliminate hazards before people are exposed.

### KEY PRINCIPLE

Prevention should begin at design and planning stage. PPE is normally the last line of defence, not the first.

### 1.3 Development of OSH in Malaysia

Malaysia moved from a prescriptive factory-and-machinery model toward a broader self-regulatory OSH system. The modern framework places responsibility on those who create, manage and perform work. The Occupational Safety and Health Act 1994 remains the principal Act, now substantially strengthened by the 2022 amendments that took effect on 1 June 2024.

| Period / instrument                               | Key significance                                                                                                                                   |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Factories and Machinery Act 1967                  | Historically controlled factories and specified machinery; repealed as part of the 2024 OSH legal restructuring.                                   |
| Occupational Safety and Health Act 1994 (Act 514) | Introduced a broader self-regulatory framework and general duties.                                                                                 |
| OSH (Amendment) Act 2022 - effective 1 Jun 2024   | Expanded duties and coverage; introduced risk-assessment duty, OSH Coordinator and employee right concerning imminent danger; increased penalties. |
| 2024 subsidiary legislation                       | Includes Construction Work (Design and Management) and Plant Requiring Certificate of Fitness regulations.                                         |
| 2025 subsidiary legislation                       | DOSH lists Occupational Safety and Health (Special Scheme of Inspection) Regulations 2025.                                                         |

### 1.4 Objectives and Philosophy of OSHA 1994

- Secure the safety, health and welfare of persons at work.
- Protect persons other than employees from safety or health risks arising from work activities.
- Promote a work environment adapted to workers physiological and psychological needs.
- Encourage a system of regulations and approved industry codes of practice working with the Act.
- Support self-regulation: those who create and control workplace risks must actively manage them.

### 1.5 Scope of the Act - Current Position

The Act applies to places of work throughout Malaysia, including public services and statutory authorities, subject to limited exclusions in the First Schedule. The listed exclusions are domestic employment, armed forces, and work on board ships governed by specified merchant shipping laws.

#### EXAM FOCUS

- Do not memorise the old limited-industry coverage. After the 2024 changes, the Act has broad workplace application.
- Know the three First Schedule exclusions at a basic level.
- Understand the meaning of self-regulation: responsibility is shared but not equal; duty follows control and influence.

### 1.6 General Duties of Employers - Section 15

An employer must, so far as is practicable, ensure the safety, health and welfare at work of employees. The duty includes safe plant and systems of work, safe handling/storage/transport of plant and substances, necessary information/instruction/training/supervision, safe workplace access and egress, a safe working environment with adequate welfare, and emergency procedures.

| Employer duty                       | Engineering example                                                |
|-------------------------------------|--------------------------------------------------------------------|
| Safe plant and systems              | Preventive maintenance, guards, interlocks and lockout procedures. |
| Safe use/handling/storage/transport | Chemical segregation, safe lifting plan, forklift routes.          |

| Employer duty                                   | Engineering example                                                                |
|-------------------------------------------------|------------------------------------------------------------------------------------|
| Information, instruction, training, supervision | Induction, machine-specific training, toolbox briefings and competent supervision. |
| Safe workplace and access/egress                | Clear walkways, guarded platforms, safe stairs and emergency exits.                |
| Safe environment and welfare                    | Ventilation, lighting, sanitation, drinking water and suitable rest facilities.    |
| Emergency procedures                            | Alarm, spill response, evacuation, rescue arrangements and drills.                 |

## 1.7 OSH Policy - Section 16

Except where prescribed otherwise, every employer must prepare and revise a written statement of general OSH policy and the organisation and arrangements for implementing it, and bring it to employees attention.

- Policy should be signed/endorsed by top management.
- State commitment to prevention, legal compliance and continual improvement.
- Define responsibilities and arrangements.
- Be communicated, understood, implemented and periodically reviewed.

## 1.8 New Duty to Conduct and Implement Risk Assessment - Section 18B

Every employer, self-employed person or principal must conduct risk assessment for safety and health risks posed to persons who may be affected by the undertaking. Where controls are needed, they must be implemented. This makes risk assessment a direct statutory duty, not merely good practice.

### LEGAL CONSEQUENCE

Contravention of sections 15, 16, 17, 18, 18A or 18B can attract a fine up to RM500,000, imprisonment up to two years, or both.

## 1.9 Duties of Employees - Sections 24 to 26

- Take reasonable care of own safety and health and that of others affected by acts or omissions.
- Co-operate with the employer or others in discharging OSH duties.
- Wear or use PPE or protective clothing provided for risk prevention.
- Comply with OSH instructions or measures instituted at work.
- Do not intentionally, recklessly or negligently interfere with or misuse safety provisions.
- Employees must not be charged for things done or provided pursuant to OSH requirements.

## 1.10 Employee Right in Imminent Danger - Section 26A

An employee who has reasonable justification to believe there is imminent danger must first inform the employer or representative. If the employer fails to take action to remove the danger, the employee has the right to remove himself or herself from the danger or work. The employee is protected against undue consequences and discrimination when the right is exercised according to the section.

### IMMINENT DANGER

The Act defines imminent danger as a serious risk of death or serious bodily injury caused by a workplace hazard such as plant, substance, condition, activity, process, practice or procedure.

## 1.11 Safety Organisation: SHO, OSH Coordinator and Committee

| Requirement                            | When / purpose                                                                                                                                                         |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Safety and Health Officer (SHO) - s.29 | Required for classes/descriptions of workplaces specified by ministerial order; competent person dedicated to compliance and promotion of safe work.                   |
| OSH Coordinator - s.29A                | Where workplace is not in a class requiring SHO, an employer with 5 or more employees must appoint an employee as OSH Coordinator, unless an SHO is already appointed. |
| Safety & Health Committee - s.30       | Required where 40 or more persons are employed, or where directed by the Director General. Supports consultation, review and resolution of OSH matters.                |

## 1.12 Important Subsidiary Legislation to Recognise

| Regulation                                                                                                                 | Why it matters to students                                                        |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Safety and Health Committee Regulations 1996                                                                               | Committee composition and operation.                                              |
| Control of Industrial Major Accident Hazards Regulations 1996 (CIMAH)                                                      | Major accident hazards involving dangerous substances.                            |
| Use and Standards of Exposure of Chemicals Hazardous to Health Regulations 2000 (USECHH)                                   | Chemical health risk assessment, exposure control and health-related duties.      |
| Notification of Accident, Dangerous Occurrence, Occupational Poisoning and Occupational Disease Regulations 2004 (NADOPOD) | Workplace event notification and records.                                         |
| Classification, Labelling and Safety Data Sheet of Hazardous Chemicals Regulations 2013 (CLASS)                            | GHS-aligned classification, labels and SDS.                                       |
| Noise Exposure Regulations 2019                                                                                            | Noise risk, assessment and control.                                               |
| Construction Work (Design and Management) Regulations 2024                                                                 | Safety and health duties across construction design and management.               |
| Plant Requiring Certificate of Fitness Regulations 2024                                                                    | Current framework for prescribed plant requiring certificates of fitness.         |
| Special Scheme of Inspection Regulations 2025                                                                              | Current DOSH-listed subsidiary legislation concerning special inspection schemes. |

## 1.13 Practicable vs Reasonably Practicable

The Act frequently uses the phrase “so far as is practicable”. In exams, explain that duty holders are expected to take feasible preventive measures considering the severity of risk, available knowledge, availability and suitability of controls, and cost in relation to risk. High-severity foreseeable hazards normally justify stronger controls.

### EXAM FOCUS

- Be able to compare employer duties (s.15) with employee duties (s.24).
- Memorise s.18B = risk assessment and implementation of controls.
- Memorise s.26A = right to remove from imminent danger after informing employer and employer fails to act.
- Memorise s.29A = OSH Coordinator; 5 or more employees (subject to the section).
- Memorise s.30 = committee at 40 or more employees or DG direction.

## CHAPTER 2

# OCCUPATIONAL SAFETY & HEALTH MANAGEMENT

Turning legal duties into a functioning safety management system

## 2.1 What is OSH Management?

OSH management is the organised process of setting policy, identifying hazards, assessing risks, allocating responsibilities, implementing controls, checking performance and improving the system. A mature system treats safety as part of normal business and engineering decision-making.

## 2.2 The Plan-Do-Check-Act (PDCA) Cycle

| Stage | Typical OSH activities                                                                                                              |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|
| PLAN  | Policy, legal register, objectives, responsibilities, hazard identification, risk assessment, control planning, emergency planning. |
| DO    | Implement controls, safe work procedures, training, communication, contractor control, maintenance, PPE and emergency readiness.    |
| CHECK | Inspection, monitoring, exposure measurement, incident investigation, audits, KPI review and corrective actions.                    |
| ACT   | Management review, lessons learned, update controls, revise objectives, improve resources and system design.                        |

### EXAM SHORTCUT

PDCA is a useful structure for any “explain how an organisation manages OSH” question. Give 2-3 examples under each stage.

## 2.3 Elements of an Effective OSH Policy

- Clear commitment from top management.
- Commitment to prevent work-related injury and ill health.
- Commitment to comply with applicable legal and other requirements.
- Framework for OSH objectives and measurable targets.
- Clear allocation of responsibilities and authority.
- Worker consultation and participation.
- Commitment to continual improvement.
- Communication and periodic review.

## 2.4 Roles and Responsibilities

| Role                      | Core responsibilities                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Top management            | Set policy, provide resources, assign responsibilities, lead by example, review performance.                            |
| Line manager / supervisor | Implement safe systems, conduct briefings, correct unsafe conditions, supervise work and respond to reports.            |
| Engineer / designer       | Eliminate hazards by design, specify safe equipment, consider maintainability, access, guarding and fail-safe features. |
| SHO / OSH Coordinator     | Coordinate OSH activities, advise, monitor, support risk assessment, training and reporting within legal role.          |
| Safety & Health Committee | Consult, review measures, investigate concerns, recommend improvements and support cooperation.                         |
| Employee                  | Follow procedures, use PPE, report hazards/incidents, cooperate and take reasonable care.                               |
| Contractor                | Comply with site requirements, assess own work risks, provide competent persons and coordinate interfaces.              |
| Visitor                   | Follow site instructions, stay within authorised areas and use required PPE.                                            |

## 2.5 Safety and Health Committee

A committee is a formal channel for employer-employee consultation. It is not a substitute for management responsibility. Its value comes from regular review of actual workplace conditions and follow-up of corrective actions.

- Review OSH measures and their effectiveness.
- Investigate unsafe or unhealthy matters raised at the workplace.
- Attempt to resolve concerns and escalate where necessary.
- Review incidents, trends, inspection findings and training needs.
- Record decisions, persons responsible and target dates.

## 2.6 Planning: Objectives, Targets and Programmes

| Weak objective    | Better SMART objective                                                                                                      |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Improve safety.   | Reduce overdue machine-guarding corrective actions to zero by 30 Nov 2026.                                                  |
| Reduce accidents. | Reduce hand injury frequency in Workshop A by 30% within 12 months through guarding, tool redesign and competency training. |
| Train staff.      | Achieve 100% induction completion before employees begin unsupervised workshop work.                                        |

## 2.7 Training, Competency and Supervision

Training is effective only when it changes capability and behaviour. Identify competency requirements by task and risk. High-risk work may require formal competency, authorisation or registered competent persons under legislation.

- Induction: site hazards, emergency routes, reporting and basic rules.
- Task training: machine operation, chemical handling, lifting, electrical safety.
- Refresher training: after changes, incidents, poor performance or lapse of time.
- Verification: observation, practical assessment, quizzes and authorisation.
- Supervision: intensity should match worker experience, complexity and risk.

## 2.8 Communication and Worker Participation

- Toolbox talks and pre-job briefings.
- Safety alerts after incidents or near misses.
- Notice boards and digital communication.
- Hazard/near-miss reporting channels.
- Consultation during risk assessment and change management.
- Stop-work / escalation mechanisms consistent with legal rights.

## 2.9 Inspection vs Audit

| Inspection                                    | Audit                                                                                                 |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Looks at workplace conditions and activities. | Evaluates whether the management system is designed and implemented effectively.                      |
| Often frequent: daily/weekly/monthly.         | Usually periodic and planned.                                                                         |
| Examples: guard missing, spill, blocked exit. | Examples: risk assessments not reviewed, training process ineffective, corrective-action system weak. |
| Immediate corrective actions common.          | System-level findings and management actions common.                                                  |

## 2.10 Leading and Lagging Indicators

| Leading indicators (preventive)   | Lagging indicators (outcomes) |
|-----------------------------------|-------------------------------|
| Risk assessments completed        | Lost-time injuries            |
| Corrective actions closed on time | Medical treatment cases       |
| Safety observations / inspections | Occupational disease cases    |
| Training competency completion    | Property damage               |
| Preventive maintenance compliance | Days lost / injury rates      |

### GOOD PRACTICE

Do not manage safety only by “zero accident” numbers. A workplace can report zero injuries while serious uncontrolled hazards remain. Combine leading and lagging indicators.

## 2.11 Emergency Preparedness and Response

1. Identify credible emergencies: fire, chemical spill, gas release, electrical incident, structural failure, severe weather, confined-space emergency.
2. Define alarms, command structure, communication and evacuation/rescue arrangements.
3. Provide equipment: extinguishers, spill kits, emergency showers/eyewash where applicable, first-aid resources.
4. Train relevant personnel and conduct drills.
5. Coordinate with external emergency services where necessary.
6. After drills/incidents, review performance and improve the plan.

## 2.12 Management of Change (MOC)

Changes can create new hazards. Before introducing a new machine, chemical, layout, software control, process parameter, contractor or work method, evaluate the change, update risk assessments and procedures, train affected persons and verify controls before normal operation.

**EXAM FOCUS**

- Know PDCA and give examples.
- Differentiate inspection and audit.
- Differentiate leading and lagging indicators.
- Explain why worker participation improves hazard identification.
- For management questions, link policy -> planning -> implementation -> checking -> improvement.

## CHAPTER 3

# INCIDENT PREVENTION, INVESTIGATION & REPORTING

Learning before the same event happens again

## 3.1 Accident, Incident and Near Miss

| Term                           | Simple meaning                                                      | Example                                                            |
|--------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|
| Accident                       | Unplanned event that results in injury, ill health, damage or loss. | Worker receives a cut from unguarded sheet metal.                  |
| Incident                       | Work-related event in which harm occurred or could have occurred.   | Forklift collision with a rack; no injury.                         |
| Near miss                      | Incident with no actual harm but meaningful potential for harm.     | Load drops beside worker.                                          |
| Dangerous occurrence           | Specified serious event that may require statutory notification.    | Certain collapse, explosion or plant failure events as prescribed. |
| Occupational poisoning/disease | Work-related poisoning or disease as prescribed.                    | Chemical poisoning or occupational dermatitis.                     |

## 3.2 Why Incidents Happen: Immediate and Root Causes

An unsafe act or unsafe condition may be the immediate cause, but effective prevention requires investigation of deeper system causes. Blaming the injured person usually stops the investigation too early.

| Level                | Examples                                                                                                                                                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Immediate causes     | Bypassed guard, slippery floor, wrong tool, exposed conductor, poor body position.                                                                         |
| Contributing factors | Fatigue, time pressure, unclear instruction, poor lighting, inadequate supervision.                                                                        |
| Root / system causes | Poor design standard, weak procurement, absent maintenance strategy, ineffective training system, poor change management, unrealistic production planning. |

## 3.3 Accident Causation Models

- **Domino concept:** A sequence of weaknesses can lead to an incident; prevention removes or controls key links.
- **Swiss-cheese model:** Multiple preventive barriers have weaknesses. Incidents occur when weaknesses align, allowing a hazard to pass through several layers.
- **Energy model:** Harm occurs when uncontrolled energy (mechanical, electrical, thermal, chemical, pressure, gravity, radiation) reaches a person or asset.
- **Human factors model:** Performance is shaped by task design, workload, competence, environment, equipment interface and organisational conditions.

### 3.4 Unsafe Acts and Unsafe Conditions

| Unsafe act / behaviour        | Unsafe condition                           |
|-------------------------------|--------------------------------------------|
| Removing a machine guard      | Missing or defective guard                 |
| Using damaged electrical tool | Frayed cable / exposed conductor           |
| Walking under suspended load  | Poorly controlled lifting zone             |
| Not using required PPE        | No suitable PPE available / poor selection |
| Speeding with forklift        | Poor route design / no segregation         |

#### IMPORTANT

Do not use “human error” as the final root cause. Ask why the error was possible, likely or not detected.

### 3.5 Hierarchy of Controls

| Priority           | Control                 | Example                                                                    |
|--------------------|-------------------------|----------------------------------------------------------------------------|
| 1 - Most effective | Elimination             | Remove the task, hazard or dangerous substance entirely.                   |
| 2                  | Substitution            | Use a less hazardous chemical or lower-energy process.                     |
| 3                  | Engineering controls    | Guarding, enclosure, isolation, local exhaust ventilation, automation.     |
| 4                  | Administrative controls | Procedure, permit, training, signs, job rotation, scheduling, supervision. |
| 5 - Last line      | PPE                     | Helmet, gloves, goggles, respirator, hearing protection.                   |

### 3.6 Incident Investigation Objectives

- Provide care and make the area safe first.
- Preserve evidence where safe and legally appropriate.
- Determine what happened, where, when and who was involved.
- Identify immediate, contributing and root causes.
- Determine failed or missing controls.
- Recommend corrective and preventive actions.
- Share lessons with relevant work groups.
- Verify actions are implemented and effective.

### 3.7 Investigation Process

- Initial response and scene control.
- Collect evidence: photographs, sketches, equipment state, documents, CCTV where applicable.
- Interview witnesses separately using open questions.
- Develop event timeline.
- Identify hazards and energy sources.
- Analyse causes using methods such as 5 Whys, barrier analysis or fishbone diagram.
- Develop actions using hierarchy of controls.
- Assign responsible person and due date.
- Follow up and verify effectiveness.

### 3.8 The 5 Whys - Worked Example

| Question                              | Example answer                                                                                       |
|---------------------------------------|------------------------------------------------------------------------------------------------------|
| Why was the operator hand cut?        | Hand contacted moving cutting blade.                                                                 |
| Why could the hand contact the blade? | Guard was removed.                                                                                   |
| Why was the guard removed?            | Material frequently jammed and operators needed access.                                              |
| Why did material jam?                 | Feed mechanism was worn and misaligned.                                                              |
| Why was it not corrected?             | Preventive maintenance did not include alignment checks and repeated jam reports were not escalated. |

Root-level action: repair/redesign the feed mechanism, restore interlocked guarding, improve preventive maintenance and defect escalation. “Tell operator to be careful” is not enough.

### 3.9 Reporting and NADOPD 2004

The Occupational Safety and Health (Notification of Accident, Dangerous Occurrence, Occupational Poisoning and Occupational Disease) Regulations 2004 establish notification requirements for prescribed workplace events. In practice, organisations should have an internal reporting system that triggers legal review so that reportable events are notified to DOSH within the applicable requirements.

#### STUDY NOTE

For examinations, focus on the categories: accident, dangerous occurrence, occupational poisoning and occupational disease; the purpose is prompt notification, investigation, records and prevention. For actual workplace compliance, always check the current regulation and DOSH instructions rather than relying on memory.

### 3.10 Corrective Action Quality

| Weak action                   | Stronger action                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------|
| Remind workers to be careful. | Install interlocked guard and verify it prevents operation when open.                                         |
| Conduct toolbox talk.         | Redesign pedestrian-forklift segregation and add physical barriers; briefing supports the engineering change. |
| Wear gloves.                  | Eliminate sharp edge / deburr automatically; select gloves only for residual risk.                            |
| Put warning sign.             | Fix leak source, improve containment and floor drainage; sign is temporary/supporting control.                |

### 3.11 Behaviour, Culture and Just Culture

A strong safety culture encourages reporting, learning and accountability. Deliberate violations and reckless behaviour require management, but unintentional errors should trigger examination of task design and system conditions. Fear-based reporting systems hide near misses and remove opportunities to learn.

#### EXAM FOCUS

- Differentiate immediate cause and root cause.
- Use hierarchy of controls when proposing corrective action.
- Be able to write a simple 5 Whys chain.
- Explain why near-miss reporting is valuable.
- Know that investigation aims to prevent recurrence, not simply allocate blame.



## CHAPTER 4

# CHEMICAL SAFETY & HIRARC

Recognising hazards, assessing risk and selecting effective controls

## 4.1 Types of Workplace Hazards

| Hazard category     | Examples                                                                                                   |
|---------------------|------------------------------------------------------------------------------------------------------------|
| Physical            | Noise, vibration, heat, cold, radiation, poor lighting, pressure.                                          |
| Chemical            | Solvents, acids, alkalis, gases, dusts, fumes, vapours, sensitizers.                                       |
| Biological          | Bacteria, viruses, mould, blood/body-fluid exposure.                                                       |
| Mechanical / safety | Moving machinery, sharp edges, falls, vehicles, stored energy.                                             |
| Electrical          | Live conductors, damaged insulation, arc flash, poor earthing.                                             |
| Ergonomic           | Heavy lifting, awkward posture, repetition, force, prolonged standing.                                     |
| Psychosocial        | Excessive workload, violence, bullying, low control, poor role clarity, fatigue-related work organisation. |

## 4.2 Chemical Routes of Entry

| Route              | Explanation                                | Control examples                                                                              |
|--------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------|
| Inhalation         | Breathing gas, vapour, dust, mist or fume. | Substitution, enclosure, local exhaust ventilation, respiratory protection for residual risk. |
| Skin / eye contact | Splash or direct contact.                  | Closed transfer, splash guards, gloves, goggles/face shield, hygiene.                         |
| Ingestion          | Contaminated hands, food or smoking.       | No eating in chemical areas, handwashing, hygiene.                                            |
| Injection          | High-pressure equipment or puncture.       | Pressure isolation, guarding, safe needle/sharp handling.                                     |

## 4.3 CLASS Regulations 2013 and GHS Communication

The CLASS Regulations 2013 govern classification, labelling and Safety Data Sheets (SDS) for hazardous chemicals. The purpose is to communicate chemical hazards consistently so users know the hazard class, precautions, emergency measures and safe handling requirements.

| Label / SDS element     | Purpose                                                 |
|-------------------------|---------------------------------------------------------|
| Product identifier      | Identifies the chemical/product.                        |
| Signal word             | Indicates relative severity, e.g. Danger or Warning.    |
| Hazard pictogram        | Visual symbol for hazard type.                          |
| Hazard statement        | Standardised description of the hazard.                 |
| Precautionary statement | Prevention, response, storage and disposal precautions. |
| Supplier information    | Identifies responsible supplier.                        |

| Label / SDS element | Purpose                                                              |
|---------------------|----------------------------------------------------------------------|
| Safety Data Sheet   | Detailed 16-section safety and health information for workplace use. |

#### 4.4 The 16 SDS Sections - Know the Structure

| No. | SDS section                      | What you look for                                   |
|-----|----------------------------------|-----------------------------------------------------|
| 1   | Identification                   | Product, use, supplier, emergency contact           |
| 2   | Hazard identification            | Classification, pictograms, signal word, statements |
| 3   | Composition / ingredients        | Hazardous ingredients and concentration             |
| 4   | First-aid measures               | Immediate response                                  |
| 5   | Fire-fighting measures           | Extinguishing media and fire hazards                |
| 6   | Accidental release measures      | Spill precautions and cleanup                       |
| 7   | Handling and storage             | Safe practices and incompatibilities                |
| 8   | Exposure controls / PPE          | Exposure limits, ventilation and PPE                |
| 9   | Physical and chemical properties | Flash point, pH, vapour pressure etc.               |
| 10  | Stability and reactivity         | Incompatible materials and conditions               |
| 11  | Toxicological information        | Health effects and exposure routes                  |
| 12  | Ecological information           | Environmental effects                               |
| 13  | Disposal considerations          | Safe disposal guidance                              |
| 14  | Transport information            | Transport classification                            |
| 15  | Regulatory information           | Applicable regulatory information                   |
| 16  | Other information                | Revision and additional information                 |

#### 4.5 USECHH Regulations 2000 - Core Concept

The USECHH Regulations 2000 focus on protection from chemicals hazardous to health at work. Key concepts include identifying hazardous chemicals, assessing health risks, controlling exposure, maintaining controls, providing information/training, and where applicable exposure monitoring, health surveillance and medical surveillance by competent persons.

#### 4.6 HIRARC: Hazard Identification, Risk Assessment and Risk Control

HIRARC is a structured method for identifying hazards from work activities, estimating risk and deciding controls. It supports the statutory duty under s.18B of OSHA 1994 to conduct risk assessment and implement necessary controls.

16. Define activity / process and persons exposed.
17. Identify hazards and possible consequences.
18. Identify existing controls.
19. Estimate likelihood (L).
20. Estimate severity (S).
21. Calculate or determine risk rating (commonly L x S in a 5 x 5 matrix).
22. Decide whether risk is acceptable/tolerable or requires action.
23. Select additional controls using hierarchy of controls.
24. Assign owner and target date.
25. Reassess residual risk after controls and review when conditions change.

#### 4.7 Example 5 x 5 Risk Matrix

| Severity \<br>Likelihood | 1 Rare | 2 Unlikely | 3 Possible | 4 Likely | 5 Almost certain |
|--------------------------|--------|------------|------------|----------|------------------|
| 5 Catastrophic           | 5 M    | 10 H       | 15 H       | 20 E     | 25 E             |

| Severity \ Likelihood | 1 Rare | 2 Unlikely | 3 Possible | 4 Likely | 5 Almost certain |
|-----------------------|--------|------------|------------|----------|------------------|
| 4 Major               | 4 M    | 8 M        | 12 H       | 16 E     | 20 E             |
| 3 Moderate            | 3 L    | 6 M        | 9 M        | 12 H     | 15 H             |
| 2 Minor               | 2 L    | 4 M        | 6 M        | 8 M      | 10 H             |
| 1 Insignificant       | 1 L    | 2 L        | 3 L        | 4 M      | 5 M              |

Illustrative interpretation used in these notes: L = Low, M = Medium, H = High, E = Extreme. Organisations may use different matrices and criteria; always follow the approved workplace method.

## 4.8 Likelihood and Severity Guide

| Likelihood       | Typical interpretation              | Severity        | Typical consequence                      |
|------------------|-------------------------------------|-----------------|------------------------------------------|
| 1 Rare           | Exceptional / very unlikely         | 1 Insignificant | No injury or negligible impact           |
| 2 Unlikely       | Could occur but not expected        | 2 Minor         | First-aid level injury / minor effect    |
| 3 Possible       | Might occur sometimes               | 3 Moderate      | Medical treatment / temporary disability |
| 4 Likely         | Expected in many circumstances      | 4 Major         | Serious injury / permanent disability    |
| 5 Almost certain | Repeated/frequent exposure or event | 5 Catastrophic  | Fatality or multiple severe outcomes     |

## 4.9 Worked HIRARC Example - Bench Grinder

| Activity / hazard                    | Possible consequence            | Existing control            | L | S | Risk | Additional control                                                                                                          |
|--------------------------------------|---------------------------------|-----------------------------|---|---|------|-----------------------------------------------------------------------------------------------------------------------------|
| Grinding - wheel fragments / contact | Eye/face injury, laceration     | Basic guard, safety glasses | 3 | 4 | 12 H | Inspect wheel, correct tool rest, tongue guard, face shield, training, prohibit side grinding unless wheel designed for it. |
| Grinding - sparks                    | Burn / ignition of combustibles | Housekeeping                | 3 | 3 | 9 M  | Spark containment, remove combustibles, hot-work controls where relevant.                                                   |
| Noise                                | Hearing damage                  | Short exposure              | 3 | 3 | 9 M  | Assess noise exposure; engineering noise control; hearing protection programme if required.                                 |
| Entanglement                         | Serious hand/body injury        | Operator awareness          | 2 | 4 | 8 M  | No loose clothing/jewellery; secure hair; guarding; training and supervision.                                               |

## 4.10 Risk Control Selection

Choose controls that act on the hazard rather than relying only on worker behaviour. Combine controls when necessary. After implementation, verify that the control does not create a new hazard.

| Control level  | Chemical example                                  | Mechanical example                                       |
|----------------|---------------------------------------------------|----------------------------------------------------------|
| Eliminate      | Remove solvent-cleaning step                      | Design out manual cutting task                           |
| Substitute     | Water-based cleaner for more hazardous solvent    | Lower-speed/low-energy process                           |
| Engineering    | Closed transfer + LEV                             | Interlocked guard / enclosure                            |
| Administrative | SOP, permit, restricted access                    | Training, inspection, maintenance schedule               |
| PPE            | Chemical goggles/gloves/respirator as appropriate | Face shield, gloves only where safe, protective footwear |

## 4.11 Review Triggers for Risk Assessment

- New process, machine, chemical or layout.
- Incident, near miss, occupational disease or unexpected exposure.
- Legal or standard change.
- Control failure or inspection/audit finding.
- Change in production rate, staffing, work hours or contractor arrangement.
- New information about hazard severity.
- Periodic scheduled review according to organisation requirements.

## 4.12 Permit-to-Work (PTW)

A permit-to-work is a formal control for selected high-risk non-routine tasks. It confirms hazards, isolations, precautions, responsible persons and validity period. Common applications include hot work, confined space, electrical isolation, excavation and certain work at height. A permit does not make unsafe work safe by itself; the specified controls must actually be implemented and verified.

## 4.13 Lockout/Tagout and Energy Isolation

26. Identify all energy sources: electrical, hydraulic, pneumatic, mechanical, thermal, pressure, gravity.
27. Shut down equipment normally.
28. Isolate each energy source.
29. Apply lock and identification tag according to authorised procedure.
30. Release or restrain stored energy.
31. Verify zero-energy state before work.
32. Perform work.
33. Inspect area and remove tools/personnel.
34. Authorised persons remove locks and restore energy under procedure.

### EXAM FOCUS

- Hazard = potential source of harm; risk = likelihood + severity.
- Know the five hierarchy levels in order.
- Be able to calculate  $L \times S$  and explain the risk level using a given matrix.
- For HIRARC questions, always state existing control AND additional control.
- Know why SDS is used and recognise important sections 2, 4, 5, 6, 7, 8 and 10.

## CHAPTER 5

# FIRE SAFETY, EMERGENCY RESPONSE & FIRST AID

Prevent ignition, control fire growth and protect people

## 5.1 Fire Triangle and Fire Tetrahedron

A fire requires fuel, oxygen and heat. The fire tetrahedron adds the chemical chain reaction that sustains combustion. Fire control works by removing or interrupting one or more of these elements.

| Fire element    | Control idea                                                                         |
|-----------------|--------------------------------------------------------------------------------------|
| Fuel            | Good housekeeping, limit flammable inventory, close containers, control leaks.       |
| Oxygen          | Inerting or smothering where appropriate; normally not a general workplace strategy. |
| Heat / ignition | Control hot work, electrical faults, static, smoking, hot surfaces and friction.     |
| Chain reaction  | Certain extinguishing agents interrupt combustion chemistry.                         |

## 5.2 Common Sources of Workplace Fire

- Electrical overload, damaged wiring and poor connections.
- Hot work: welding, cutting, grinding.
- Flammable liquid/vapour release.
- Gas leaks.
- Hot surfaces and process heaters.
- Static electricity.
- Smoking and open flames.
- Poor housekeeping and accumulation of combustible waste.
- Incompatible chemical storage.

## 5.3 Fire Classes and Extinguishing Media

| Class / fuel                     | Examples                              | Common suitable media*                                          |
|----------------------------------|---------------------------------------|-----------------------------------------------------------------|
| A - ordinary combustibles        | Wood, paper, cloth, many plastics     | Water, foam, dry powder depending on situation.                 |
| B - flammable liquids            | Petrol, solvents, paint               | Foam, dry powder, CO2 depending on fire and equipment.          |
| C - flammable gases              | LPG, methane                          | Dry powder; shut off gas supply when safe and trained to do so. |
| D - combustible metals           | Magnesium, sodium, some metal powders | Special Class D powder matched to metal.                        |
| Electrical equipment (energised) | Switchboards, motors, appliances      | CO2 or suitable dry powder; isolate power when possible.        |
| F - cooking oils/fats            | Deep-fat fryers                       | Wet chemical extinguisher.                                      |

\*Exact extinguisher selection must follow site fire risk assessment, equipment approval and local requirements. Never use water on energised electrical equipment or incompatible reactive materials.

## 5.4 Extinguisher Use - PASS as a Memory Aid

| P | Pull the pin                |
|---|-----------------------------|
| A | Aim at the base of the fire |
| S | Squeeze the handle          |
| S | Sweep side to side          |

### WHEN NOT TO FIGHT A FIRE

Do not attempt firefighting if you are not trained, the fire is spreading, smoke is heavy, the correct extinguisher is unavailable, hazardous chemicals/gases are involved beyond your competence, or your escape route is not secure. Raise the alarm, evacuate and call emergency services.

## 5.5 Fire Prevention Programme

- Control ignition sources and hot work.
- Maintain electrical systems and avoid overloading.
- Store flammable liquids/gases correctly.
- Separate incompatible chemicals.
- Maintain housekeeping and waste removal.
- Keep fire doors, exits and access routes clear.
- Inspect and maintain fire protection systems.
- Train fire wardens / emergency response personnel as applicable.
- Conduct evacuation drills and review performance.

## 5.6 Emergency Evacuation

35. Recognise alarm and stop work safely if possible.
36. Proceed by the designated escape route; do not delay to collect belongings.
37. Do not use lifts unless the emergency plan specifically permits a protected evacuation lift.
38. Assist others only within training and without creating additional casualties.
39. Report to the assembly point.
40. Supervisors/wardens account for persons using the site system.
41. Do not re-enter until authorised.

## 5.7 Emergency Plan Information

| Plan element  | Questions to answer                                                       |
|---------------|---------------------------------------------------------------------------|
| Alarm         | How is alarm raised? Are signals distinguishable?                         |
| Roles         | Who is incident controller, warden, first aider, spill responder?         |
| Routes        | Where are primary/alternate exits?                                        |
| Assembly      | Where do people report and how is accountability done?                    |
| Communication | Who contacts 999, security, management and neighbouring areas?            |
| Special risks | Chemicals, gas cylinders, high voltage, confined space, disabled persons? |
| Recovery      | Who authorises re-entry and incident investigation?                       |

## 5.8 Basic First-Response Priorities

First aid is immediate assistance given before professional medical care. Training matters: students should use this section for theory, not as a substitute for certified first-aid training. In a serious emergency in Malaysia, activate the site emergency system and call 999.

42. Check for danger to yourself, the casualty and others. Do not enter an unsafe area.
43. Check response and summon help.
44. Assess airway and breathing according to current first-aid training.
45. If there is life-threatening bleeding, control it using trained first-aid methods and appropriate protective barriers.
46. Do not move a casualty unnecessarily where spinal or serious injury is suspected, unless remaining in place creates greater danger.
47. Provide information to emergency responders and record the incident according to workplace procedure.

## 5.9 Common Workplace First-Aid Situations - High-Level Response

| Situation                   | Priority actions (summary only)                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minor cut                   | Protect yourself, control bleeding, clean/dress according to training, report.                                                                                             |
| Severe bleeding             | Call emergency help, apply direct pressure using appropriate barrier, monitor until responders arrive.                                                                     |
| Chemical splash to eye/skin | Use emergency eyewash/shower immediately as indicated by SDS/site procedure; remove contaminated clothing when safe; seek medical assessment.                              |
| Burn                        | Remove from heat source when safe; cool thermal burn with cool running water according to current first-aid guidance; do not apply inappropriate creams or break blisters. |
| Electric shock              | Do not touch casualty until electrical source is isolated; call emergency help and provide trained response.                                                               |
| Inhalation exposure         | Do not enter contaminated atmosphere without appropriate protection; move to fresh air only if safe; call emergency assistance.                                            |
| Suspected fracture          | Keep casualty still and supported; seek medical help; avoid unnecessary movement.                                                                                          |

## 5.10 Chemical Spill Response

48. Identify substance if this can be done safely; consult SDS/spill procedure.
49. Raise alarm and isolate area.
50. Remove ignition sources only if safe and authorised.
51. Use correct PPE and spill kit only if trained and the spill is within local response capability.
52. Prevent spread to drains/environment where safe and permitted.
53. Escalate major/unknown spills to emergency responders.
54. Dispose of cleanup materials as required and investigate the release.

## 5.11 Emergency Drills: What to Evaluate

- Alarm heard/understood?
- Evacuation time and route problems?
- Blocked doors or congestion?

- Accountability accurate?
- Wardens clear about roles?
- Emergency communication effective?
- Persons needing assistance supported?
- Any unsafe behaviour during drill?
- Corrective actions assigned and closed?

**EXAM FOCUS**

- Fire triangle = heat + fuel + oxygen; tetrahedron adds chain reaction.
- Know Class A/B/C/D/F and suitable general extinguisher concepts.
- PASS is a memory aid, but evacuation comes first when conditions are unsafe.
- For emergency questions: alarm -> safe shutdown -> evacuate -> assembly/accountability -> emergency services -> review.
- First aid: protect rescuer, summon help, follow trained procedures; never create a second casualty.

## CHAPTER 6

# WORKPLACE ENVIRONMENT & ERGONOMICS

Controlling exposure and fitting work to people

## 6.1 Workplace Environment

A healthy work environment controls physical, chemical, biological, ergonomic and psychosocial exposures. Many occupational illnesses develop gradually, so the absence of immediate injury does not mean the workplace is safe.

## 6.2 Noise

Excessive occupational noise can cause permanent hearing damage and can also interfere with communication and alarms. The OSH (Noise Exposure) Regulations 2019 form part of the current Malaysian legal framework for workplace noise.

- Identify noisy processes and persons exposed.
- Assess noise using competent methods where required.
- Prioritise quieter equipment, enclosure, isolation, damping and maintenance.
- Use administrative controls to support engineering controls.
- Provide suitable hearing protection for residual exposure and ensure correct fit/use.
- Where required, implement hearing conservation measures such as information/training and audiometric testing.

## 6.3 Lighting

| Problem          | Potential effect                                                   | Control                                               |
|------------------|--------------------------------------------------------------------|-------------------------------------------------------|
| Too little light | Errors, eye strain, trips, poor inspection quality                 | Increase general/task lighting; maintain luminaires.  |
| Glare            | Eye discomfort, headaches, poor screen visibility                  | Reposition lights/screens; shielding; matte surfaces. |
| Flicker          | Discomfort; can create misleading perception of rotating machinery | Maintain/replace lighting; suitable drivers/fixtures. |
| High contrast    | Visual fatigue                                                     | Balance task and surrounding illumination.            |

## 6.4 Thermal Environment

Heat stress depends on air temperature, humidity, radiant heat, air movement, workload, clothing and individual factors. Controls should reduce heat at source and workload exposure before relying only on hydration advice.

| Control type | Examples                                                       |
|--------------|----------------------------------------------------------------|
| Engineering  | Insulation, shielding, ventilation, local cooling, automation. |

| Control type   | Examples                                                                                     |
|----------------|----------------------------------------------------------------------------------------------|
| Administrative | Work-rest schedule, acclimatisation, scheduling heavy work in cooler periods, buddy system.  |
| Personal       | Appropriate clothing/cooling PPE where suitable, access to drinking water and recovery area. |

## 6.5 Ventilation and Indoor Air Quality

- General/dilution ventilation supplies and removes air to control background contaminants and comfort.
- Local exhaust ventilation (LEV) captures contaminants near the source before they enter the breathing zone.
- LEV components include hood, duct, air cleaner, fan and discharge.
- Poor maintenance, blocked ducts or incorrect hood placement can make LEV ineffective.
- Indoor air concerns may include inadequate fresh air, moisture/mould, dust, combustion products and contaminants from processes or materials.

## 6.6 Vibration

| Type                 | Examples                | Possible effects                             | Controls                                                                |
|----------------------|-------------------------|----------------------------------------------|-------------------------------------------------------------------------|
| Hand-arm vibration   | Grinders, impact tools  | Vascular/neurological symptoms, reduced grip | Low-vibration tools, maintenance, exposure management, task redesign.   |
| Whole-body vibration | Forklift, heavy vehicle | Back discomfort and fatigue                  | Seat/suspension, route maintenance, speed control, exposure management. |

## 6.7 Ergonomics: Fitting Work to the Worker

Ergonomics designs tasks, tools, equipment and environments to match human capabilities and limitations. The goal is not simply “good posture”; it is to reduce excessive force, repetition, awkward posture, contact stress, static loading and poorly designed work organisation.

## 6.8 Musculoskeletal Disorder (MSD) Risk Factors

| Risk factor     | Examples                                      | Design response                                                               |
|-----------------|-----------------------------------------------|-------------------------------------------------------------------------------|
| Force           | Heavy lifting, forceful grip, pushing/pulling | Reduce load, mechanical aid, better handles, powered assistance.              |
| Repetition      | Fast repeated assembly                        | Automation, task redesign, recovery time, reduce unnecessary motions.         |
| Awkward posture | Bending, twisting, overhead work              | Adjustable height, reposition parts/tools, access redesign.                   |
| Static posture  | Prolonged standing/sitting                    | Movement variation, workstation adjustment, sit-stand options where suitable. |
| Contact stress  | Sharp edge pressing on wrist/leg              | Round/pad edges, change support surface.                                      |
| Vibration       | Powered hand tools                            | Low-vibration tools, maintenance, exposure reduction.                         |

## 6.9 Manual Handling - TILE Memory Aid

| T - Task        | Twisting? Reaching? Long carry? High frequency? Need team lift?                |
|-----------------|--------------------------------------------------------------------------------|
| I - Individual  | Capability, training, fatigue, pregnancy/health considerations where relevant. |
| L - Load        | Weight, size, shape, stability, grip, sharp/hot contents.                      |
| E - Environment | Space, floor, stairs, lighting, temperature, obstacles.                        |

### BETTER THAN “LIFT CORRECTLY”

First ask whether the manual lift can be eliminated, mechanised or reduced. Training in lifting technique is an administrative control and should not replace good task design.

## 6.10 Basic Workstation Ergonomics

- Frequently used items within easy reach.
- Work surface height suited to task and worker; precision work may need different height from forceful work.
- Monitor positioned to reduce sustained neck flexion/rotation.
- Keyboard/mouse arranged to minimise excessive wrist deviation and reach.
- Chair provides suitable support and adjustability where seated work is required.
- Provide leg/foot space and avoid sharp contact points.
- Encourage task variation and movement rather than one fixed “perfect posture”.

## 6.11 Prolonged Standing

DOSH has published Guidelines on Occupational Safety and Health for Standing at Work 2024. Prolonged standing should be assessed as an ergonomic exposure. Controls can include redesigning tasks to allow sitting or movement, adjustable work heights, suitable footwear/flooring, anti-fatigue solutions where appropriate, and work-rest/task variation based on risk assessment.

## 6.12 Psychosocial Risk

Psychosocial hazards arise from how work is designed, organised and managed. DOSH published Guidelines on Psychosocial Risk Assessment and Management at the Workplace 2024, reflecting that occupational health includes psychological as well as physical well-being.

| Psychosocial hazard     | Examples of organisational controls                                     |
|-------------------------|-------------------------------------------------------------------------|
| Excessive demands       | Realistic staffing/workload, workload review, prioritisation.           |
| Low control             | Worker input into methods/schedules where feasible.                     |
| Poor support            | Supervisor capability, access to assistance, clear escalation channels. |
| Role ambiguity/conflict | Clear responsibilities, priorities and reporting lines.                 |
| Bullying/violence       | Prevention policy, reporting, investigation and support mechanisms.     |
| Fatigue / long hours    | Scheduling, rest, staffing and fatigue risk controls.                   |

## 6.13 Occupational Hygiene Control Strategy

55. Anticipate hazards during design and planning.
56. Recognise hazards and exposed groups.

57. Evaluate exposure using qualitative and quantitative methods.
58. Control exposure using hierarchy of controls.
59. Verify controls through inspection, monitoring, health information and maintenance.
60. Review when processes, standards or evidence change.

## 6.14 Engineering Design for Safety and Ergonomics

- Design access for safe maintenance.
- Locate valves, controls and displays within safe reach and visibility.
- Provide lifting points for heavy components.
- Reduce need to enter hazardous zones for routine inspection.
- Use fail-safe/interlocked control where appropriate.
- Design drains, containment and isolation for spills/maintenance.
- Consider human-machine interface, alarms and error-proofing.

### EXAM FOCUS

- Noise: control at source/path before PPE.
- LEV captures contaminant at/near source; general ventilation dilutes background contamination.
- Ergonomics = fit work to people, not force people to adapt to poor design.
- Manual handling: eliminate/mechanise first; technique training alone is weak.
- Recognise psychosocial hazards as OSH hazards and propose organisational controls.

## CHAPTER A

# HIRARC WORKED EXAMPLE & TEMPLATE

A complete example you can adapt for tutorials

## A.1 Worked HIRARC - Engineering Workshop

| Work activity           | Hazard / event                             | Consequence                           | Existing controls              | L | S | Initial risk | Additional control                                                                                       | Residual risk |
|-------------------------|--------------------------------------------|---------------------------------------|--------------------------------|---|---|--------------|----------------------------------------------------------------------------------------------------------|---------------|
| Lathe operation         | Entanglement with rotating chuck/workpiece | Severe injury / fatality              | Fixed guards; trained operator | 2 | 5 | 10 H         | Interlocked chuck guard; prohibit loose clothing; secure hair; pre-use check; supervised competency      | 5 M           |
| Manual lifting of motor | Heavy/awkward load                         | Back/shoulder injury                  | Two-person lift when available | 3 | 3 | 9 M          | Use trolley/hoist; lifting points; reduce storage height; team lift only for residual cases              | 3 L           |
| Solvent cleaning        | Vapour + skin exposure                     | Irritation / CNS effects / dermatitis | Gloves; open tray              | 3 | 3 | 9 M          | Substitute less hazardous cleaner; closed parts washer/LEV; correct glove selection from SDS             | 3 L           |
| Electrical testing      | Contact with live conductors               | Electric shock / burn                 | Basic insulated tools          | 2 | 5 | 10 H         | De-energised testing where possible; lockout; guarded test points; authorised persons; rated instruments | 5 M           |
| Grinding                | Flying particles                           | Eye/face injury                       | Safety glasses                 | 3 | 4 | 12 H         | Wheel inspection, correct guard/tool rest, face shield, training, maintenance                            | 4 M           |
| Forklift movement       | Vehicle-pedestrian collision               | Crush injury / fatality               | Marked route                   | 3 | 5 | 15 H         | Physical segregation, controlled crossings, speed control, visibility aids, traffic plan                 | 5 M           |

## A.2 Blank HIRARC Template

| No. | Activity | Hazard | Consequence | Existing controls | L | S | Risk | Additional controls | Person / due date | Residual risk |
|-----|----------|--------|-------------|-------------------|---|---|------|---------------------|-------------------|---------------|
| 1   |          |        |             |                   |   |   |      |                     |                   |               |
| 2   |          |        |             |                   |   |   |      |                     |                   |               |
| 3   |          |        |             |                   |   |   |      |                     |                   |               |
| 4   |          |        |             |                   |   |   |      |                     |                   |               |
| 5   |          |        |             |                   |   |   |      |                     |                   |               |
| 6   |          |        |             |                   |   |   |      |                     |                   |               |

## A.3 HIRARC Answering Formula for Exams

61. State the work activity.
62. Name the hazard specifically (not just “unsafe”).
63. State credible consequence.
64. State existing controls.
65. Assign likelihood and severity using the given table.
66. Calculate risk score.
67. Propose controls in hierarchy order.
68. Reassess residual risk if requested.

### COMMON ERROR

Students often list “PPE” first. Higher marks usually come from proposing elimination, substitution or engineering controls when these are reasonably possible, then adding administrative controls/PPE for residual risk.

## CHAPTER B

# INCIDENT INVESTIGATION CHECKLIST

A structured method for case-study questions

## B.1 Scene and Evidence Checklist

- Make safe and provide emergency care.
- Record date, time, exact location and activity.
- Photograph overall scene and close-up details where authorised.
- Record machine/control positions and energy state.
- Identify environmental conditions: lighting, weather, noise, housekeeping.
- Secure relevant parts/equipment when required.
- Collect permits, risk assessments, SOPs, training records and maintenance records.
- Identify witnesses and supervisors.

## B.2 Interview Questions

- What task was being performed?
- What happened immediately before the event?
- What did you see/hear/feel?
- What was different from normal?
- What instructions/training were provided?
- Were there time pressures, equipment problems or previous similar events?
- What controls were expected and were they working?
- What would make this task safer?

## B.3 Root Cause Categories

| Category                 | Questions                                                                       |
|--------------------------|---------------------------------------------------------------------------------|
| Design / engineering     | Could the hazard have been eliminated or physically prevented?                  |
| Equipment / maintenance  | Was equipment suitable, maintained and inspected?                               |
| Procedure / planning     | Was the work method adequate and practical?                                     |
| Competence / supervision | Did people have the skill, authority and supervision needed?                    |
| Communication            | Were hazards and changes communicated?                                          |
| Work organisation        | Fatigue, staffing, workload, conflicting priorities?                            |
| Management system        | Did audits, reporting and corrective-action systems detect the problem earlier? |

## B.4 Corrective Action Tracker

| Finding / cause | Action | Control level | Owner | Due date | Status / verification |
|-----------------|--------|---------------|-------|----------|-----------------------|
|                 |        |               |       |          |                       |

| Finding / cause | Action | Control level | Owner | Due date | Status / verification |
|-----------------|--------|---------------|-------|----------|-----------------------|
|                 |        |               |       |          |                       |
|                 |        |               |       |          |                       |
|                 |        |               |       |          |                       |

## CHAPTER C

## RAPID EXAM REVISION

High-yield facts, answer structures and practice questions

## C.1 One-Page Memory Map

| Topic           | Remember                                                                                                                       |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
| OSHA 1994       | Current Act 514 framework; major Amendment Act changes effective 1 Jun 2024.                                                   |
| Employer duty   | s.15 - ensure safety, health and welfare so far as practicable.                                                                |
| Risk assessment | s.18B - conduct risk assessment and implement necessary risk controls.                                                         |
| Employee duty   | s.24 - reasonable care, cooperate, use PPE, comply with instructions.                                                          |
| Imminent danger | s.26A - after informing employer; right to remove if employer fails to act; protection from undue consequences/discrimination. |
| OSH Coordinator | s.29A - many workplaces with 5+ employees not otherwise in SHO class; appoint employee as coordinator.                         |
| Committee       | s.30 - 40+ persons or DG direction.                                                                                            |
| Hierarchy       | Eliminate -> Substitute -> Engineering -> Administrative -> PPE.                                                               |
| HIRARC          | Activity -> hazard -> consequence -> existing control -> L -> S -> risk -> additional control -> residual risk.                |
| Incident        | Investigate causes and barriers; do not stop at blame.                                                                         |
| Fire            | Triangle: heat, fuel, oxygen; tetrahedron adds chain reaction.                                                                 |
| Ergonomics      | Fit work to people; reduce force, repetition, awkward/static posture, vibration and contact stress.                            |

## C.2 How to Answer “Explain” Questions

Use P-E-E: Point -> Explanation -> Example. For a 10-mark question, give several distinct points and develop each briefly. Avoid one-word lists when the command word is “explain”.

| Weak                 | Better                                                                                                                                                                        |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engineering control. | Engineering control physically separates people from the hazard. Example: an interlocked guard prevents access to a rotating machine while it is operating.                   |
| Training.            | Training builds task competency, but because it relies on human behaviour it should support stronger controls rather than replace them where engineering control is feasible. |

## C.3 Common Comparison Questions

| Compare               | Key difference                                                                   |
|-----------------------|----------------------------------------------------------------------------------|
| Hazard vs risk        | Hazard = potential source of harm; risk = likelihood/severity of harm occurring. |
| Accident vs near miss | Accident causes harm/loss; near miss does not, but had potential.                |

| Compare                               | Key difference                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Inspection vs audit                   | Inspection checks conditions/activities; audit evaluates the management system.                                                   |
| Leading vs lagging KPI                | Leading measures preventive activity; lagging measures outcomes.                                                                  |
| Engineering vs administrative control | Engineering changes physical system; administrative changes how work is organised/performed.                                      |
| Safety vs occupational health         | Safety often addresses acute injury events; occupational health includes exposure-related and long-latency illness. Both overlap. |

## C.4 Practice Short-Answer Questions

69. Define hazard and risk and give one engineering example of each.
70. State four general duties of an employee under section 24 of OSHA 1994.
71. Explain the new duty under section 18B.
72. When is an OSH Coordinator required under section 29A?
73. When must a safety and health committee be established under section 30?
74. List the hierarchy of controls in order of effectiveness.
75. Explain two reasons why PPE is considered the last line of defence.
76. Differentiate an accident, incident and near miss.
77. Give three immediate causes and three root/system causes for a machine injury.
78. Explain the purpose of an SDS and name four sections you would use before handling a chemical.
79. Calculate risk when L=4 and S=5 using a multiplication matrix.
80. Explain how local exhaust ventilation differs from general ventilation.
81. List four ergonomic risk factors for musculoskeletal disorders.
82. Explain fire triangle and fire tetrahedron.
83. State the main steps in an incident investigation.

## C.5 Practice Structured Questions with Answer Guides

Q1. A student is using a bench grinder. The tool rest is too far from the wheel, the eye shield is missing, and sparks are landing near cardboard boxes. Prepare a HIRARC-style answer.

- Hazards: contact/jamming at wheel gap, flying fragments, sparks/ignition, noise.
- Consequences: laceration/entanglement, eye/face injury, fire, hearing damage.
- Controls: correct tool-rest gap and guards, wheel inspection, eye/face protection for residual risk, remove combustibles, spark control, noise assessment/control, training and supervision.
- Use the matrix supplied in the question; do not invent a risk category if no matrix is given.

Q2. A worker reports a strong smell from a leaking chemical line. The supervisor tells the worker to continue production. Explain the correct OSH approach.

- Stop and assess the hazard; identify chemical using label/SDS.
- Isolate the leak/area and use emergency/spill procedure.
- Prevent unprotected exposure and ignition if relevant.
- Employer must assess and control risk; s.18B reinforces this duty.
- If the situation meets the statutory definition of imminent danger, s.26A provides an employee right after informing the employer and the employer fails to act.
- Investigate why the leak occurred and implement permanent control.

Q3. Explain how an organisation can reduce forklift-pedestrian collision risk.

- Eliminate unnecessary shared routes / redesign layout.
- Engineering: physical barriers, separate doors, controlled crossings, speed-limiting technology, visibility aids.
- Administrative: traffic plan, one-way rules, authorised drivers, delivery scheduling, pedestrian rules, inspection.
- PPE such as high-visibility clothing supports but does not replace segregation.
- Monitor near misses and revise traffic controls.

## C.6 Common Mistakes That Lose Marks

- Writing “hazard = risk”.
- Giving PPE as the only control for a high-risk hazard.
- Calling “carelessness” the root cause without analysing the system.
- Using old OSHA coverage information from before the 2024 amendments.
- Forgetting employee right under s.26A and risk-assessment duty under s.18B.
- Confusing a safety committee (40+ persons / DG direction) with OSH Coordinator (5+ employees in the s.29A situation).
- Memorising a risk matrix without reading the matrix supplied in the examination.
- Treating training as an engineering control.

## CHAPTER D

## GLOSSARY &amp; REFERENCES

Key terms and authoritative sources used for this edition

## D.1 Glossary

| Term                   | Meaning                                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ALARP / tolerable risk | Concept of reducing risk to a level that is as low as reasonably practicable / tolerable within the governing framework. |
| Competent person       | Person registered or recognised to perform specified OSH activities under applicable law.                                |
| Consequence            | Outcome or severity of harm if a hazardous event occurs.                                                                 |
| Control measure        | Action or barrier used to eliminate or reduce risk.                                                                      |
| Hazard                 | Source, situation or act with potential to cause harm.                                                                   |
| HIRARC                 | Hazard Identification, Risk Assessment and Risk Control.                                                                 |
| Incident               | Work-related event in which injury/ill health occurred or could have occurred.                                           |
| LEV                    | Local exhaust ventilation; captures airborne contaminants near source.                                                   |
| Likelihood             | Estimate of how probable/frequent a hazardous event is.                                                                  |
| Near miss              | Incident with no harm but potential for harm.                                                                            |
| Occupational disease   | Disease arising from or aggravated by work exposure.                                                                     |
| OSH                    | Occupational Safety and Health.                                                                                          |
| PPE                    | Personal protective equipment.                                                                                           |
| Residual risk          | Risk remaining after additional controls are implemented.                                                                |
| Risk                   | Combination of likelihood and severity/consequence of harm.                                                              |
| SDS                    | Safety Data Sheet.                                                                                                       |
| SHO                    | Safety and Health Officer.                                                                                               |
| Unsafe condition       | Physical/environmental state that increases likelihood or severity of harm.                                              |

## D.2 Authoritative Sources and Legal Update Basis

84. Department of Occupational Safety and Health (DOSH) Malaysia. Occupational Safety and Health Act 1994 (Act 514), Reprint Version 1.6.2024. Official DOSH PDF.
85. DOSH Malaysia. Regulations Under the Occupational Safety and Health Act 1994 (official legislation index; updated 16 March 2026 when accessed for this edition).
86. DOSH Malaysia. Guidance on Employees Rights for the Implementation of Section 26A of the Occupational Safety and Health Act 1994 (2024 guidance; official English publication available on DOSH portal, file republished 2026).
87. DOSH Malaysia. Guidelines on Occupational Safety and Health for Standing at Work 2024.
88. DOSH Malaysia. Guidelines on Psychosocial Risk Assessment and Management at the Workplace 2024.
89. DOSH Malaysia. Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013.
90. DOSH Malaysia. Occupational Safety and Health (Use and Standards of Exposure of Chemicals Hazardous to Health) Regulations 2000.
91. DOSH Malaysia. Occupational Safety and Health (Noise Exposure) Regulations 2019.

92. DOSH Malaysia. Occupational Safety and Health (Notification of Accident, Dangerous Occurrence, Occupational Poisoning and Occupational Disease) Regulations 2004.
93. DOSH Malaysia. Occupational Safety and Health (Construction Work) (Design and Management) Regulations 2024.
94. DOSH Malaysia. Occupational Safety and Health (Plant Requiring Certificate of Fitness) Regulations 2024.
95. DOSH Malaysia. Occupational Safety and Health (Special Scheme of Inspection) Regulations 2025.
96. Jabatan Pendidikan Politeknik dan Kolej Komuniti / POLYCC public programme materials listing DUW10042 Occupational Safety and Health for Engineering.
97. Politeknik Kuching Sarawak repository record: Simplify Notes Occupational Safety and Health for Process Engineering, Edition 2024 (used only to confirm broad topic alignment; restricted PDF content was not copied).

### D.3 Direct Official Web References

[https://dosh.gov.my/wp-content/uploads/2025/01/Occupational-Safety-and-Health-Act-1994-Act-514\\_Reprint-Version-1.6.2024\\_English.pdf](https://dosh.gov.my/wp-content/uploads/2025/01/Occupational-Safety-and-Health-Act-1994-Act-514_Reprint-Version-1.6.2024_English.pdf)

<https://dosh.gov.my/en/perundangan/peraturan/peraturan-di-bawah-akta-keselamatan-dan-kesihatan-pekerjaan-1994-akta-514/>

[https://dosh.gov.my/wp-content/uploads/2026/01/WorkersRightsHandbook\\_Translation\\_FinalPrint.pdf](https://dosh.gov.my/wp-content/uploads/2026/01/WorkersRightsHandbook_Translation_FinalPrint.pdf)

<https://repositori.mypolycc.edu.my/jspui/handle/123456789/6428>

#### LEGAL ACCURACY NOTE

OSH law can change. These notes were checked against official DOSH sources available up to 10 August 2026. For assignments or real workplace compliance, verify the latest Gazette/DOSH publication and the exact wording of the applicable Act or regulation.

### D.4 Final Study Checklist

- ☐ I can explain employer duties under s.15.
- ☐ I can explain risk assessment duty under s.18B.
- ☐ I can explain employee duties under s.24 and right under s.26A.
- ☐ I know OSH Coordinator and committee thresholds.
- ☐ I can identify hazards by category.
- ☐ I can use hierarchy of controls.
- ☐ I can complete a HIRARC table and justify controls.
- ☐ I can investigate an incident using immediate/contributing/root causes.
- ☐ I can explain SDS, chemical routes of entry and basic chemical controls.
- ☐ I can explain fire classes, evacuation and basic emergency priorities.
- ☐ I can explain noise, ventilation, heat, ergonomics and psychosocial hazards.